

Introduction

With the basics of logic under our belts, let's now apply it to classical number theory. Number theory is an interesting field of mathematics that has surprising applications to real life (think cryptography).

Learning Objectives

- Define divisibility and prove basic results using divisibility.
- State the Well-Ordering Principle and use it to prove basic results.
- Define the GCD of two numbers and prove simple results about GCDs.
- Use the Division and Euclidean Algorithms to find GCDs and find solutions to Bézout's Identity.

Divisibility

Definition 1

If $a, b \in \mathbb{Z}$, we say that a is *divisible by* b (or b divides a) if there exists some $k \in \mathbb{Z}$ such that $a = bk$. We write $b \mid a$ to indicate that b divides a .

So for example, 4 divides 12 ($4 \mid 12$) because $12 = 4 \cdot 3$. On the other hand, $5 \nmid 12$ since there is no **integer** n such that $12 = 5n$. We can also make abstract statements, such as $k \mid (k^2 + k)$, which you should convince yourself is true.

Exercise 2 For what values of n is it the case that $n \mid 0$? For what values of n is it the case that $0 \mid n$?

The following are excellent warm-up problems to try proving on your own. We will give the proofs for two of them, and leave the other three as exercise.

Proposition 3

Suppose that a, b, c are integers.

- If $a \mid b$ and $b \mid c$, then $a \mid c$.
- If $a \mid b$ and $a \mid c$, then for any $m, n \in \mathbb{Z}$ we have $a \mid (mb + nc)$.
- If $a \mid b$ and $a \mid (b + c)$, then $a \mid c$.
- If $a \mid b$ and $b \mid a$, then $a = \pm b$.
- If b is non-zero and $a \mid b$, then $|a| \leq |b|$.

Proof. We'll give here the proofs for (i) and (ii).

- Given that $a \mid b$ and $b \mid c$, we know there exist $m, n \in \mathbb{Z}$ such that $b = ma$

and $c = nb$. Thus

$$c = nb = n(ma) = ka$$

where $k = nm$ is an integer. Thus $a \mid c$ as required.

- ii. Since $a \mid b$ and $a \mid c$, let $k_1, k_2 \in \mathbb{Z}$ such that $b = k_1a$ and $c = k_2a$. Suppose that $m, n \in \mathbb{Z}$ are given arbitrarily, from which it follows that

$$mb + nc = m(k_1a) + n(k_2a) = (mk_1 + nk_2)a = \hat{k}a,$$

where $\hat{k} = mk_1 + nk_2$ is an integer. Thus $a \mid (mb + nc)$ as required. $\square$

Exercise 4 Prove Proposition 3 (iii-v).

The Euclidean Algorithm

One of the issues with a course like this is that we don't want to waste a lot of time doing things like *constructing the natural numbers* $\mathbb{N}$. We feel pretty confident that you've got a grip on what these are. Unfortunately, since we didn't construct them, there are some results that we're going to have to take as axioms; namely, results that we're going to just have to assume are true. The following is one of them:

The Well-Ordering Principle: Every non-empty subset of the positive integers has a least element. More precisely, if $S \subseteq \mathbb{Z}^+$ then there exists $m \in S$ such that $m \leq n$ for all $n \in S$.

Note that if we replace $\mathbb{Z}^+$ with $\mathbb{Z}^*$ above, the result is still true. However, replacing $\mathbb{Z}^+$ with $\mathbb{Z}$ will cause the result to be false!

The Division Algorithm

You learned about *remainders* in early primary school, but then probably didn't do too much with them. Here's where we're going to start using them again. We previously mentioned that $5 \nmid 12$, and the reason is that when we divide 12 by 5 we get a remainder term: $12 = 5 \cdot 2 + 2$. Let's prove that you can always do this.

Theorem 5: The Division Algorithm

If $a, b \in \mathbb{Z}$ with $b > 0$, there exist unique q and r such that $a = qb + r$ where $0 \leq r < b$.

Proof. Define the set $S = \{a + kb : k \in \mathbb{Z}\}$, which you should visualize as a number line where we start at a and make ticks every b -spaces.

$$S \quad \dots \quad \begin{array}{c} a-2b \\ | \\ \hline \end{array} \quad \begin{array}{c} a-b \\ | \\ \hline \end{array} \quad \begin{array}{c} a \\ | \\ \hline \end{array} \quad \begin{array}{c} a+b \\ | \\ \hline \end{array} \quad \begin{array}{c} a+2b \\ | \\ \hline \end{array} \quad \dots$$

Note that at least one element of S is non-negative. Indeed, if $a \geq 0$, then we're done. Alternatively, if $a < 0$ then $a - ab = (-a)(b - 1) \geq 0$. Hence if we define $\hat{S} = \{x \in S : x \geq 0\}$, this will be a non-empty subset of $\mathbb{Z}^*$. By the Well-Ordering Principle, $\hat{S}$ has a least element which we'll call r . Let q be the integer such that $r = a - qb$. If we can show that $0 \leq r < b$ and that both r and q are unique, then we'll have proven the theorem.

That $0 \leq r$ is immediate given that $r \in \hat{S}$. If $r \geq b$ then $0 \leq r - b = a - (q + 1)b \in \hat{S}$ which is smaller than r , which contradicts the minimality of r .

To show that r and q are unique, suppose that $\tilde{r}$ and $\tilde{q}$ are another set of integers such that $0 \leq \tilde{r} < b$ and $a = bq + r = b\tilde{q} + \tilde{r}$. For the sake of contradiction, suppose that $r \neq \tilde{r}$ and that $r > \tilde{r}$, from which it follows that $0 \leq r - \tilde{r} < b$. Additionally, we can deduce that $b(\tilde{q} - q) = r - \tilde{r}$, in which case $b \mid (r - \tilde{r})$ and thus $b < r - \tilde{r}$ from Proposition 3 (v). This contradicts the fact that $r - \tilde{r} < b$, meaning that $r = \tilde{r}$. Thus $b(\tilde{q} - q) = r - \tilde{r} = 0$, and since $b > 0$ it follows that $\tilde{q} = q$, as required. $\square$

Example 6

Show that for any $n \in \mathbb{Z}$, $3 \mid (n^3 - n)$.

Solution. There's a very fast way of doing this, but we'll leave that as an exercise. By the Division Algorithm, every number n satisfies either

$$n = 3q_0, \quad n = 3q_1 + 1, \quad \text{or} \quad n = 3q_2 + 2,$$

for some appropriate choice of $q_i \in \mathbb{Z}$. Brute forcing our way through this problem, let's plug these into $n^3 - n$ to see what we get.

In the first case where $n = 3q_0$ we get

$$n^3 - n = 27q_0^3 - 3q_0 = 3(9q_0^3 - q_0).$$

In the second case where $n = 3q_1 + 1$ we get

$$n^3 - n = 27q_1^3 + 27q_1^2 + 6q_1 = 3(9q_1^3 + 9q_1^2 + 2q_1).$$

Finally, in the third case where $n = 3q_2 + 2$ we get

$$n^3 - n = 27q_2^3 + 54q_2^2 + 33q_2 + 6 = 3(9q_2^3 + 18q_2^2 + 11q_2 + 2).$$

In all three cases, we see that $3 \mid (n^3 - n)$, demonstrating the desired result. $\blacksquare$

Exercise 7 There is a slick proof for Example 6. Can you find it? *Hint:* Try factoring $n^3 - n$.

Greatest Common Divisors

Definition 8

If $a, b \in \mathbb{Z}$, we define the *greatest common divisor* of a and b , written $\gcd(a, b)$, to be the largest positive integer that divides both a and b . More precisely,

$$\gcd(a, b) = \max \{d \in \mathbb{Z} : d \mid a \text{ and } d \mid b\}.$$

By convention, we say that $\gcd(0, 0) = 0$. The integers a and b are said to be *coprime* or *relatively prime* if $\gcd(a, b) = 1$.

For example,

$$\gcd(4, 6) = 2, \quad \gcd(15, 25) = 5, \quad \gcd(15, 33) = 3, \quad \gcd(17, 4) = 1.$$

Since every number divides 0, $\gcd(a, 0) = |a|$. If p is a prime number, then $\gcd(p, a) = 1$ unless a is a multiple of p . Note the similarity between primality and coprimality. For example, the numbers 6 and 25 are coprime since $\gcd(6, 25) = 1$, though neither number is prime on its own.

Exercise 9 Find $\gcd(1053, 481)$.

Proposition 10

If $a, b, q, r \in \mathbb{Z}$ are such that $a = bq + r$, then $\gcd(a, b) = \gcd(b, r)$.

This proposition gives us a convenient way of calculating greatest common divisors, by replacing two potentially large numbers a and b , with b and r . We know by the Division Algorithm that r can be chosen in such a way that $0 \leq r < b$, meaning that we've replaced a with a smaller number r . We can recursively continue this process until we get to a point where we can actually find the GCD. However, unlike the division algorithm, the statement that $a = bq + r$ here doesn't require q, r to be the quotient and remainder! The result will hold even if we don't use the "smallest possible" r .

Proof. Write $r = a - qb$. If $a = b = 0$ then necessarily $q = r = 0$ and so the result is trivially true. Therefore, assume that not both of a and b are zero and set $d = \gcd(a, b)$. Since $d \mid a$ and $d \mid b$ then $d \mid (a - qb) = r$ by Proposition 3(ii). Now we must show that d is in fact the greatest common divisor of r and b .

If c is any other divisor of b and r , then by Proposition 3(ii) we have $c \mid (bq + r) = a$. Since d is the greatest common divisor of a and b , it must be the case that $c \leq d$, showing that d is the greatest common divisor of b and r as required. $\square$

This result definitely looks like witchcraft the first time you see it. Spend some time thinking about it, and make sure you understand the intuition behind why it works.

Example 11

Find $\gcd(616, 427)$.

Solution. Note that $616 = 427 \cdot 1 + 189$, and so

$$\begin{aligned}
 \gcd(616, 427) &= \gcd(427, 189) && \text{since } 616 = 427 \cdot 1 + 189 \\
 &= \gcd(189, 49) && \text{since } 427 = 189 \cdot 2 + 49 \\
 &= \gcd(49, 42) && \text{since } 189 = 49 \cdot 3 + 42 \\
 &= \gcd(42, 7) && \text{since } 49 = 42 \cdot 1 + 7 \\
 &= 7.
 \end{aligned}$$

Thus by using the Division Algorithm, we reduced a complicated procedure into something algorithmic. ■

Exercise 12 If you did Exercise 9 the hard way, redo it using Proposition 10 to verify your answer.

The Euclidean Algorithm

Theorem 13: The Euclidean Algorithm

Let $a, b \in \mathbb{Z}$ with $b \neq 0$ and assume that b does not divide a . Consider the following algorithm:

$$\begin{aligned}
 a &= q_1 \underline{b} + \underline{r_1}, && \text{where } 0 < r_1 < |b| \\
 b &= q_2 \underline{r_1} + \underline{r_2}, && \text{where } 0 < r_2 < r_1 \\
 r_1 &= q_3 \underline{r_2} + \underline{r_3}, && \text{where } 0 < r_3 < r_2 \\
 &\vdots \\
 r_{n-2} &= q_n \underline{r_{n-1}} + \underline{r_n}, && \text{where } 0 < r_n < r_{n-1} \\
 r_{n-1} &= q_{n+1} \underline{r_n} + 0.
 \end{aligned}$$

This algorithm must terminate in finitely many steps, and moreover $\gcd(a, b) = r_n$.

Proof. The remainders form a strictly decreasing sequence of positive integers: $a \geq b > r_1 > r_2 > r_3 > \cdots$ and so must eventually reach zero, showing that the algorithm must eventually terminate. Furthermore, by repeatedly applying

Proposition 10 we get

$$\begin{aligned} \gcd(a, b) &= \gcd(b, r_1) = \gcd(r_1, r_2) \\ &= \gcd(r_2, r_3) \cdots \\ &= \gcd(r_{n-1}, r_n) \\ &= \gcd(r_n, 0) = r_n. \end{aligned}$$

□

This should feel as though we just made a theorem of what we did in Example 11. However, what we win with the Euclidean Algorithm is actually the ability to *undo* the process of finding the GCD. In particular, if a and b are integers with GCD $\gcd(a, b)$, then the Euclidean algorithm let's us find integers m, n such that $am + bn = \gcd(a, b)$.

Theorem 14: Bézout's Identity

If $a, b \in \mathbb{Z}$, there exist $m, n \in \mathbb{Z}$ such that $am + bn = \gcd(a, b)$.

Proof. If one of a or b is zero then the result follows quickly (why?). Thus suppose that neither a nor b is zero and define $S = \{ax + by : x, y \in \mathbb{Z}\}$. This set is non-empty since $a \in S$, and moreover it consists of at least one positive integer, since both a and $-a \in S$ and one of them is positive. Thus let $\hat{S} = \{z \in S : z > 0\}$. Since $\hat{S}$ is a non-empty set of natural numbers, by the Well-Ordering Principle it has a least element which we'll call d . Let $m, n \in \mathbb{Z}$ be such that $d = am + bn$.

We just need to show that $d = \gcd(a, b)$. Applying the Division Algorithm to a and d we can find q, r such that $a = dq + r$ with $0 \leq r < d$. But then

$$r = a - dq = a - (am + bn)q = a(1 - mq) - b(nq) \in S.$$

Since d is the smallest element of $\hat{S}$ but r is a smaller non-negative number in S , it must be the case that $r = 0$, so $d \mid a$. Since there was nothing special about the choice of a in this proof, precisely the same argument shows that $d \mid b$, and so d is a common divisor of both a and b .

To show that it is the greatest common divisor, let c be any divisor of both a and b . Fix $r, t \in \mathbb{Z}$ such that $a = cr$ and $b = ct$, from which it follows that

$$d = am + bn = (cr)m + (ct)n = c(rm + tn)$$

showing that $c \mid d$, which in turn implies that $|c| \leq d$. Thus d is the greatest common divisor of a and b . □

In fact, the proof of Bézout's Identity gives us the following two corollaries for free!

Corollary 15

If $a, b \in \mathbb{Z}$ and c is a common divisor of a and b , then $c \mid \gcd(a, b)$.

Corollary 16

If $a, b \in \mathbb{Z}$, the integer $d > 0$ is a common divisor of both a and b , and there exist $m, n \in \mathbb{Z}$ such that $am + bn = d$, then $d = \gcd(a, b)$.

Exercise 17 Convince yourself, by explicitly proving them if necessary, that Corollaries 15 and 16 are true.

Exercise 18 Suppose that $a, b \in \mathbb{Z}$ and that $d = \gcd(a, b)$. Show that

$$\gcd\left(\frac{a}{d}, \frac{b}{d}\right) = 1.$$

As mentioned, we can use the Euclidean Algorithm to find the integers m, n which occur in Bézout's Identity.

Example 19

Find an integer solution to the equation $616x + 427y = 7$.

Solution. We know from Example 11 that $\gcd(616, 427) = 7$. That solution also included the steps of the Euclidean Algorithm, which are repeated here for convenience:

$$616 = 427 \cdot 1 + 189 \quad (1)$$

$$427 = 189 \cdot 2 + 49 \quad (2)$$

$$189 = 49 \cdot 3 + 42 \quad (3)$$

$$49 = 42 \cdot 1 + 7 \quad (4)$$

$$42 = 7 \cdot 6 + 0.$$

Now we will start at the line that has our GCD (4) and begin back-substituting. In the following, we will write equations like (4) as $7 = 49 - 42 \cdot 1 = 49 + 42 \cdot (-1)$ for the sake of convenience. Square brackets will show where substitutions have been made.

$$\begin{aligned} 7 &= 49 + 42 \cdot (-1) \\ &= 49 + [189 + 49(-3)](-1) = 49(4) + 189(-1) && \text{from (3)} \\ &= [427 + 189(-2)](4) + 189(-1) = 427(4) + 189(-9) && \text{from (2)} \\ &= 427(4) + [616 + 427(-1)](-9) = 427(13) + 616(-9) && \text{from (1)} \end{aligned}$$

which one can easily double-check to verify that the result is indeed 7. ■

Exercise 20 Find an integer solution to the equation $1053x + 481y = \gcd(1053, 481)$.

Exercises

1. Suppose that $a, b, c, d \in \mathbb{Z}$. Prove the following:
 - (a) If $ac \mid bc$ and $c \neq 0$, then $a \mid b$.
 - (b) If $a \mid b$, $c \mid d$, then $ac \mid bd$.
2. Suppose that n is an odd integer. Show that $4 \mid (n^2 - 1)$.
3. For each of the pairs of numbers below, compute $\gcd(a, b)$.
 - (a) $a = 252, b = 198$
 - (b) $a = 385, b = 595$
 - (c) $a = 420, b = 378$
 - (d) $a = 4620, b = 6930$
4. Let $n \in \mathbb{Z}^+$ with $n \geq 2$. Show that for any $a \in \mathbb{Z}$ satisfying $\gcd(a, n) = 1$, there is a $b \in \mathbb{Z}$ such that $n \mid (ab - 1)$.
5. If $a, b \in \mathbb{Z}^*$, show that $\gcd(3a + 2b, a + b) = \gcd(a, b)$.